Course Summary

Danna Gurari
University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last few lectures:
  - Guest lectures from industry
- Assignments (Canvas):
  - Final project presentation and peer evaluation grades out
    - Send regrade requests to Nick Cooper
  - 5922 students: final project report due next Wednesday
- Questions?

Today's Topics

- Reflections: what we have done over the semester
- Key takeaways: lessons from the past century
- Looking ahead: how to continue with NN and DL
- Open Q&A with instructor
- Closing remarks

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Recall Course Schedule

Artificial Neurons (lecture 2)

Classical Neural Networks and Shallow Learning (lectures 3-6)

- Gradient Descent
- Fully Connected Feedforward, Recurrent, and Convolutional Neural Networks

Deep Learning and Transformers (lectures 7-19)

- Residual Learning, Batch Normalization, Hardware Acceleration, Fine Tuning, Big Data
- Applications with Visual, Textual, Audio, and Multimodal Data

Neural Networks in Society (lectures 20-26)

- Sustainable, Responsible, and Future Models
- Industry Perspectives

Three key breakthroughs:
"A spark can become a flame, a flame a fire."
- Kathryn Lasky

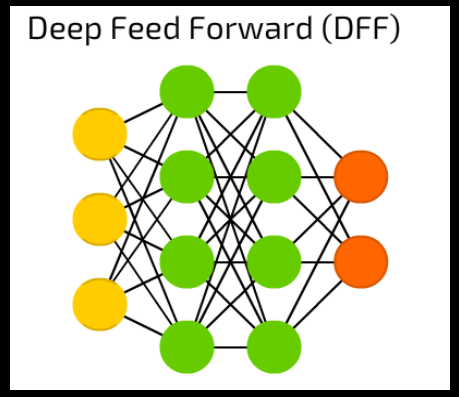


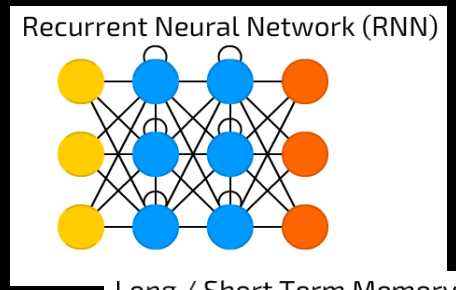


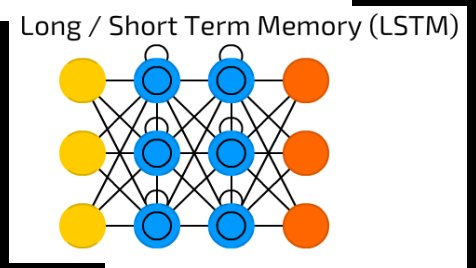


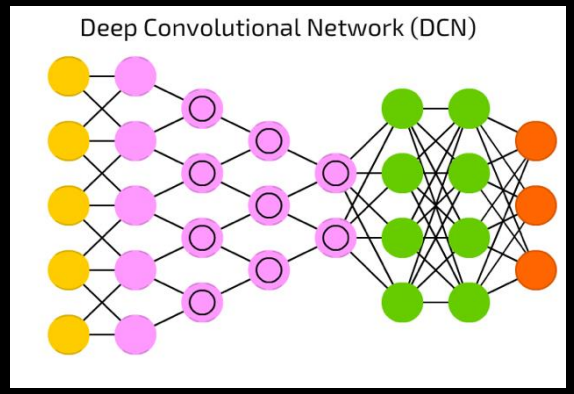
Recall Course Learning Objective 1: Recognize and Analyze Common Neural Network Architectures

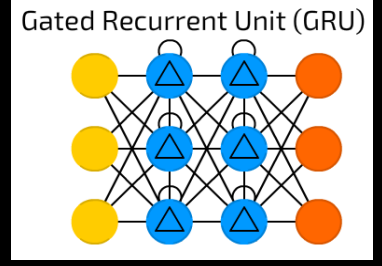
Key "Wave 2" Architectures



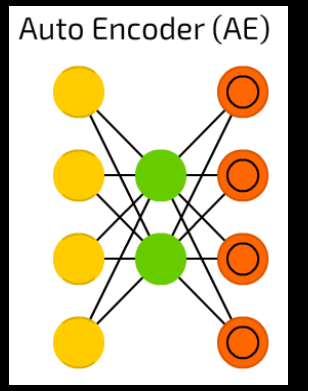


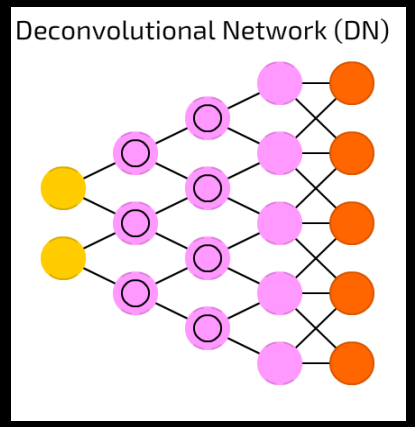


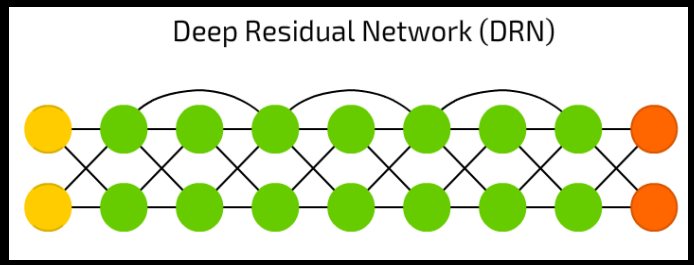


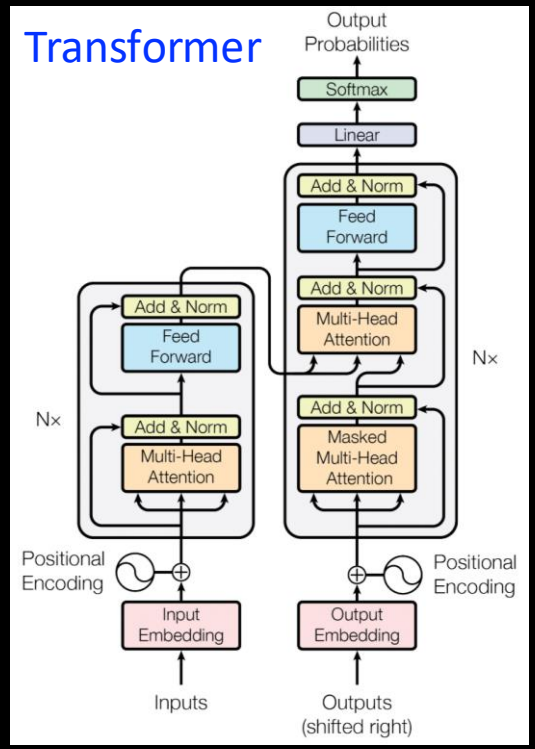


Key "Wave 3" Architectures







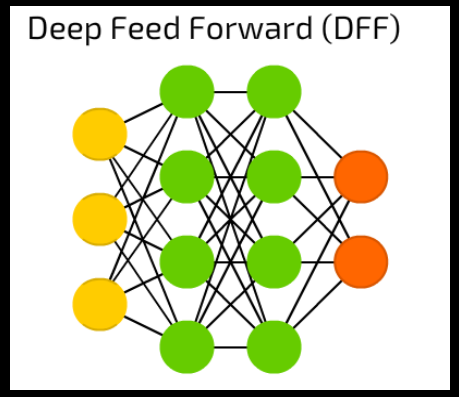


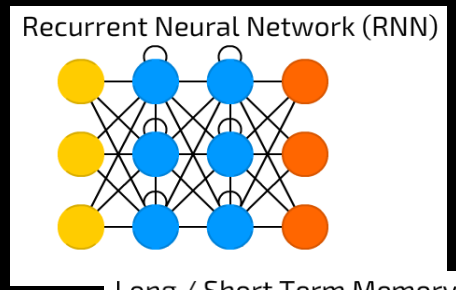
Architectures are designed to encode assumptions about data

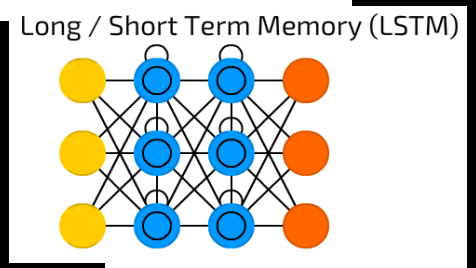
https://medium.com/data-science/the-mostly-complete-chart-of-neural-networks-explained-3fb6f2367464; Vaswani et al. Attention Is All You Need. Neurips 2017

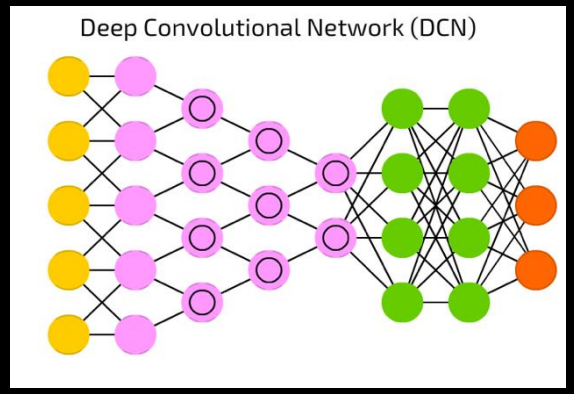
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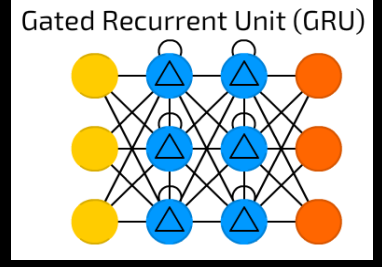
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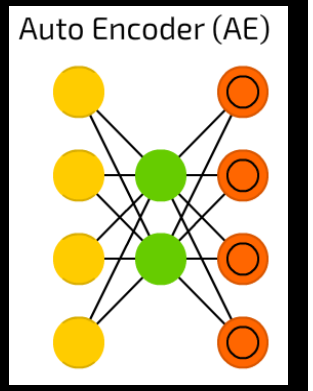


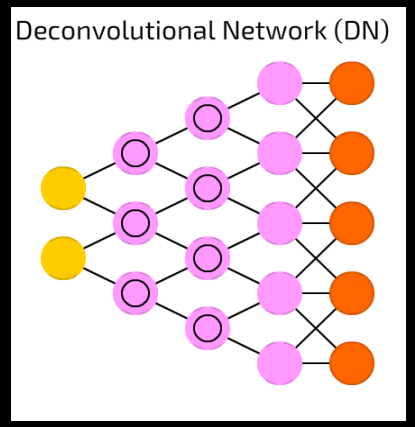


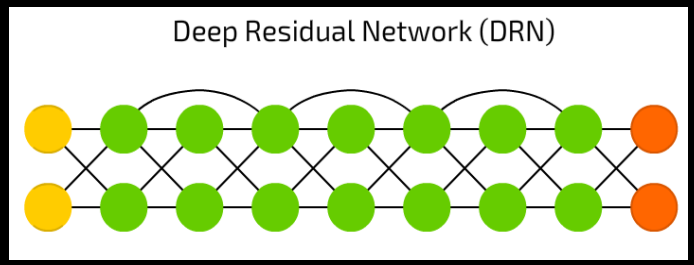


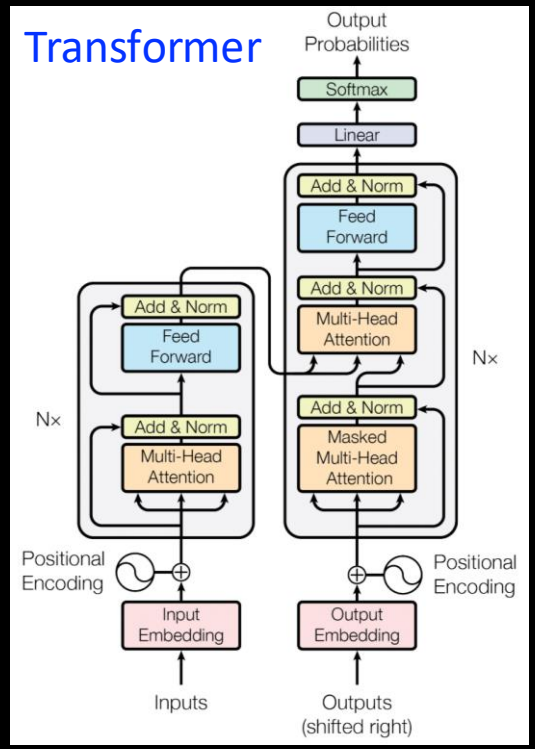


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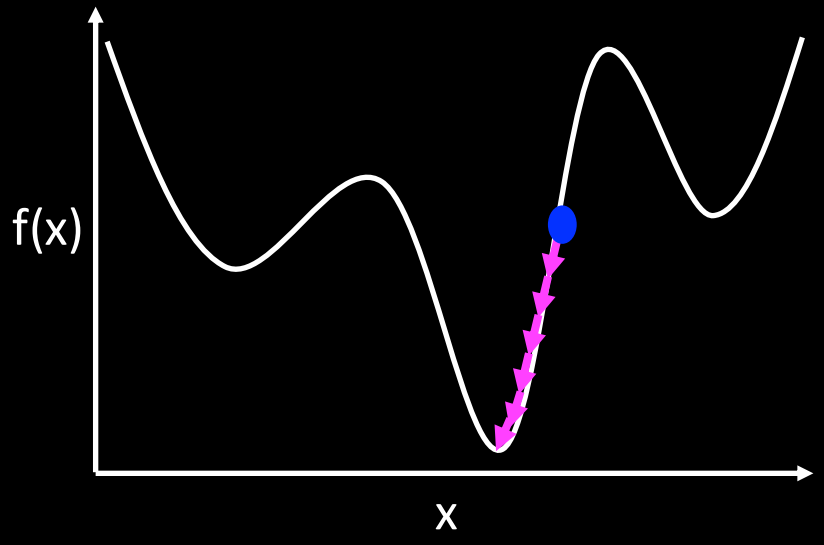


Dominant architectures changed, suggesting new one will emerge (if not yet here)

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Recall Course Learning Objective 2: Recognize and Analyze Common Techniques for Training Deep Neural Networks

Key "Wave 2" Innovations
(gradient descent, with proper initialization and optimization tricks)



Key "Wave 3" Innovations

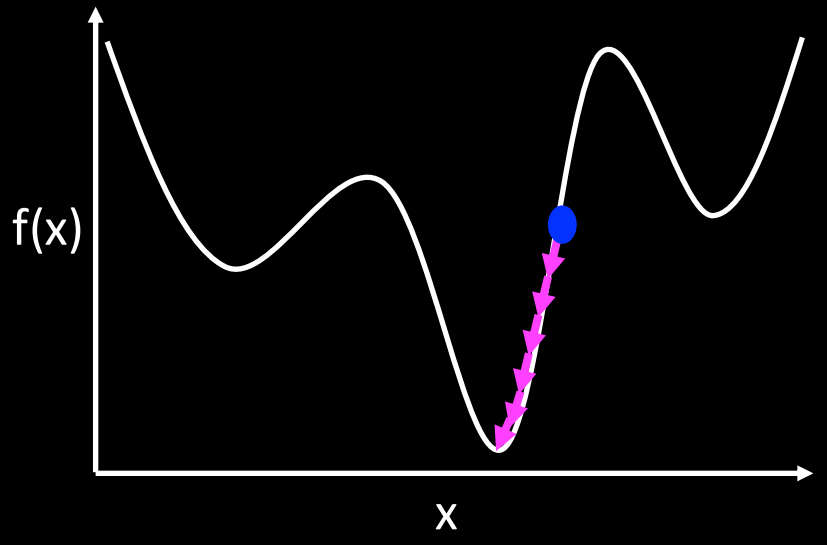
- **Task-specific** — e.g., big data, GPUs, residual learning
- **Pretraining + Fine-Tuning** — e.g., unsupervised + task-specific learning
- **Internet-Scale Training + Zero-/Few-Shot Learning**
- **Post-Training** — e.g., instruction tuning, RLHF

1986

Deep learning success = optimization breakthroughs + compute + data

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Key "Wave 2" Innovations
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Key "Wave 3" Innovations

Task-specific	Pretraining + Fine-Tuning	Internet-Scale Training + Zero-/Few-Shot Learning	Post-Training
e.g., big data, GPUs, residual learning	e.g., unsupervised + task-specific learning		e.g., instruction tuning, RLHF

1986

Dominant learning strategies changed,
suggesting new one will emerge (if not yet here)

Recall Course Objective 3: Gain Hands-on Experience with Neural Networks and Deep Learning for Various Artificial Intelligence Tasks

A. Experimented with established deep learning libraries
   - 13 coding tutorials focused mostly on PyTorch, but covered more (e.g., Hugging Face)

B. Developed deep learning models from scratch
   - 3 lab assignments covering single and multi-modality tasks , including working with "deep" architectures and "big data"

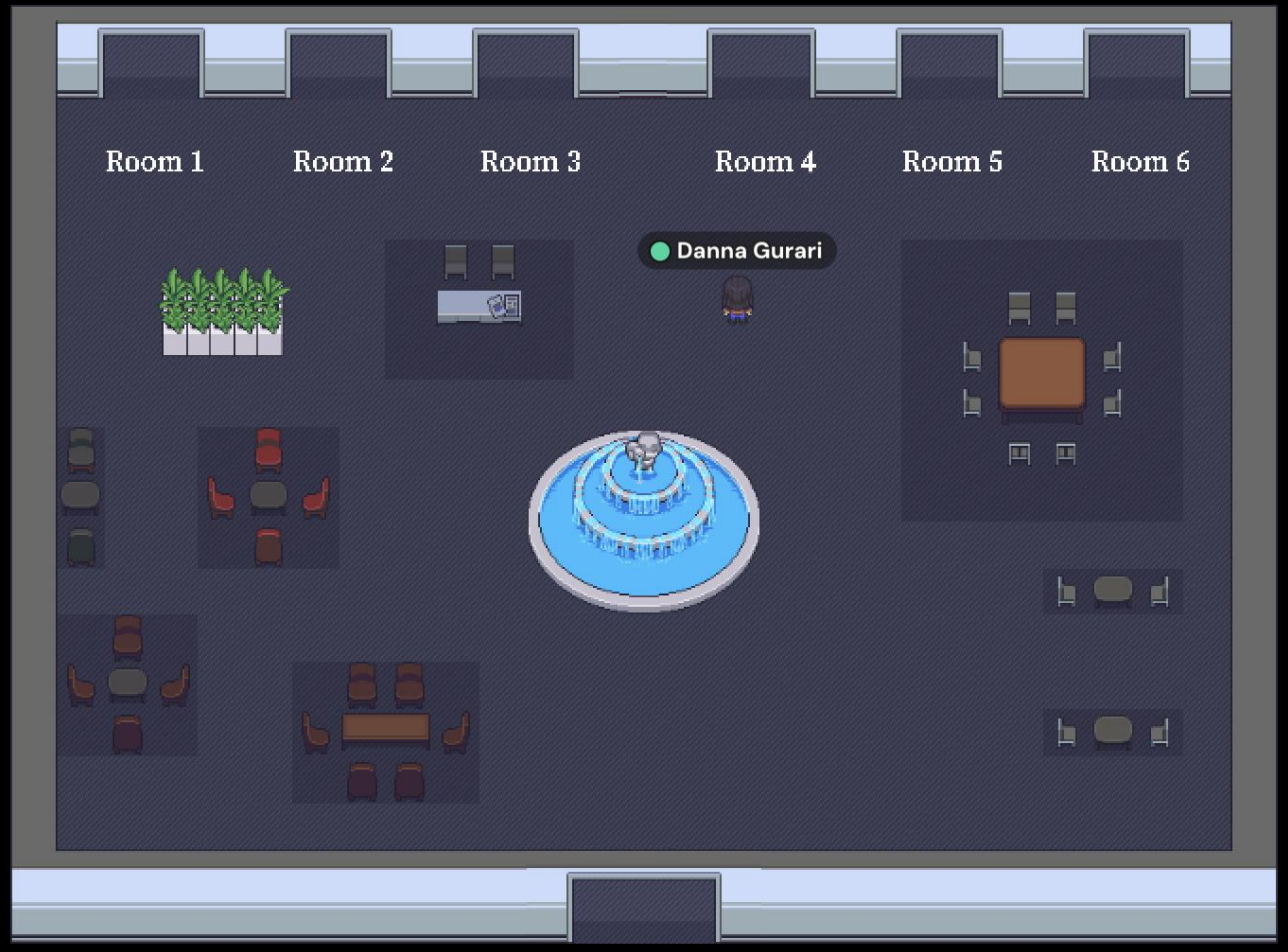
C. Evaluated deep learning models for diverse tasks, especially for analyzing text and images

Recall Course Objective 4: Explore Responsible AI Practices to Both Reduce Environmental Harm and Promote Ethical Behavior in AI Systems

2 lectures on sustainable AI with one from guest speaker (Dr. Eric Hersh)

3 lectures on AI ethics with one from guest speaker (Dr. Mehrnoosh Sameki)

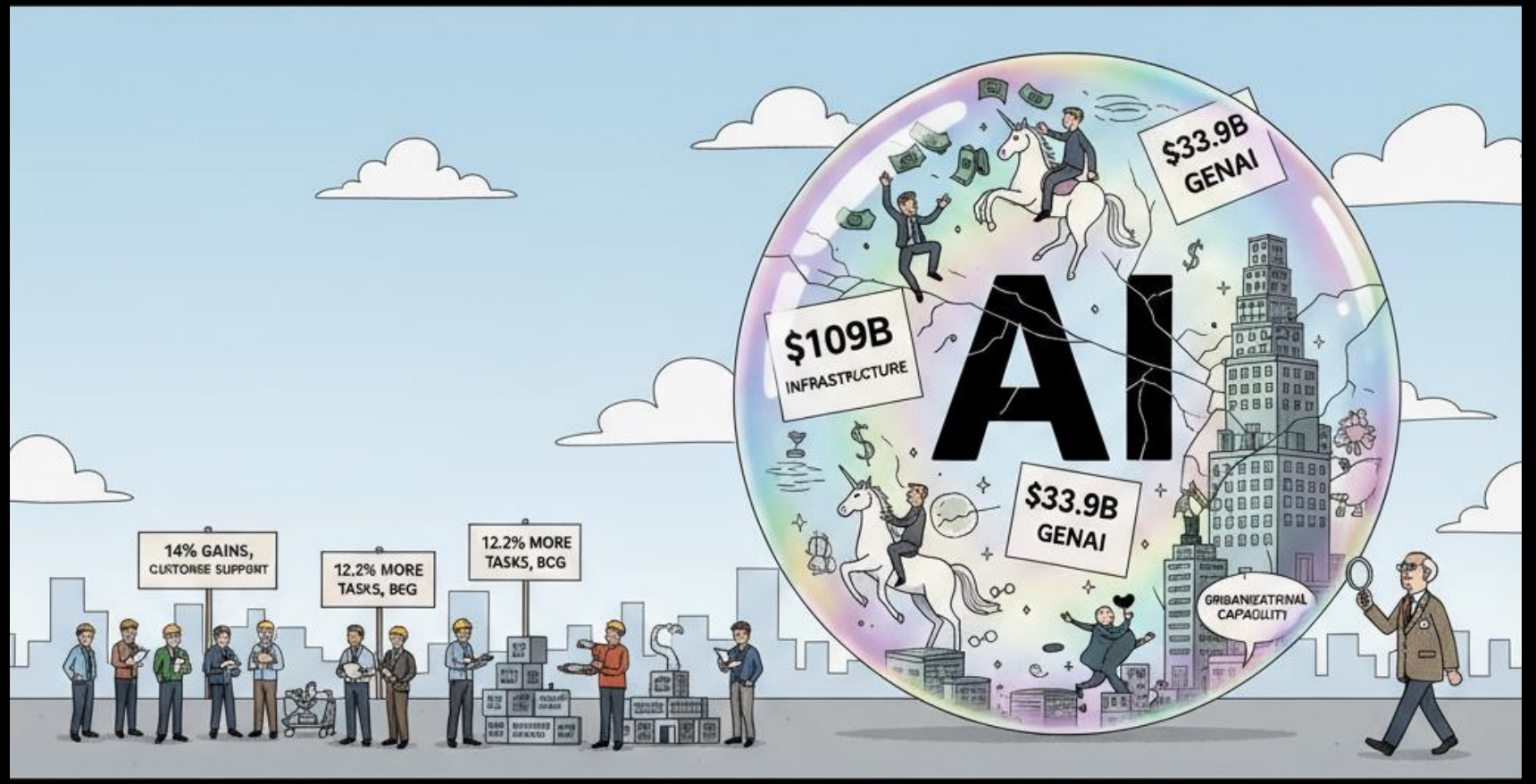
Recall Course Objective 5: Apply What is Learned to Propose and Critique Novel Projects Involving Neural Networks and Deep Learning



Today's Topics

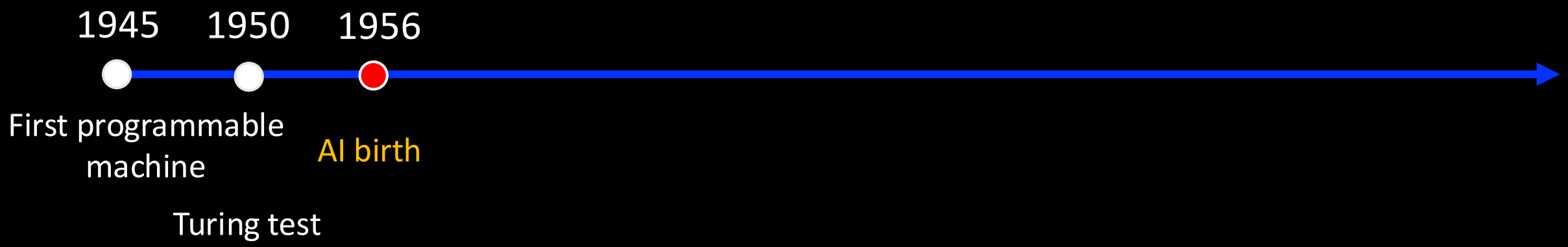
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Given History of AI, Will There be a 3rd “AI Winter”? If Yes, Will the AI Bubble Pop or Deflate?



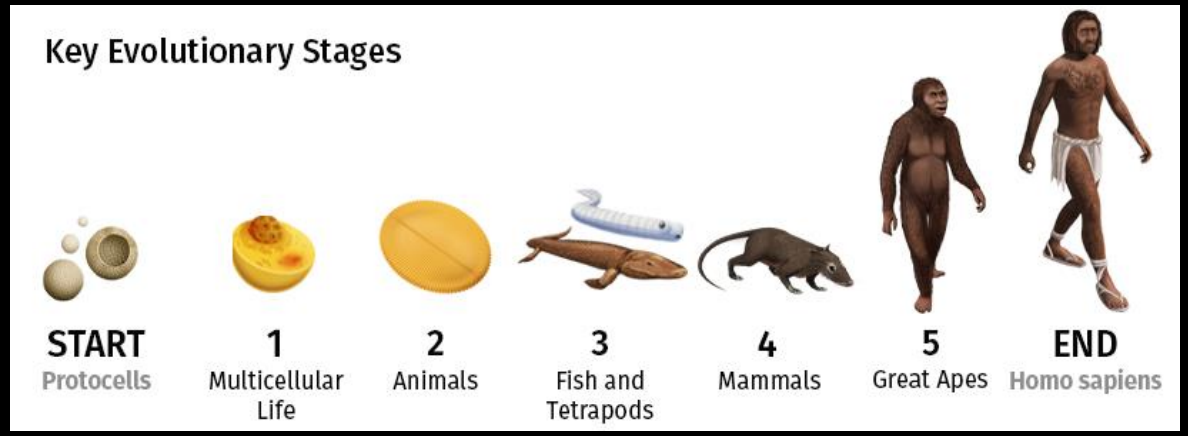
https://www.linkedin.com/pulse/we-ai-bubble-100-billion-answer-habib-baluwala-ph-d-0jqgc/

Recall AI as a Field Was Born 70 Years Ago



AI is Evolving Relatively Fast (70 Years Old) and Shaping Our Society; Context:

It took 4 billion years to evolve into modern humans!



https://www.visualcapitalist.com/path-of-human-evolution/

In that time, we continuously shaped our surroundings; e.g.,



https://www.usbr.gov/uc/rm/crsp/gc/



https://ocj.com/2023/08/2023-ohio-crop-tour-north-leg/

Open Question: How Will AI Continue to Evolve and Shape Our World?

What’s Next: Choosing How to Evolve with AI

As an educator, researcher, and mother, I champion responsible, sustainable AI that underpins a society I would want the next generation to inherit



- For you:
  - What do you find acceptable versus unacceptable from AI?
  - How will you engage with AI - Today? In 5 years? In 10 years?
  - How will you differentiate yourself from AI - Today? In 5 years? In 10 years?

- For society:
  - How will you raise awareness about AI?
  - Which AI initiatives will you fund with your money?
  - Which AI companies will you reward with your time?
  - Which government representatives will you support?

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Ways to Stay Connected with NN & DL

- Many courses at CU Boulder
- Read research papers: e.g., on arXiv and Google Scholar
- Attend workshops and conferences: e.g., best paper awardees at NeurIPS, ICML, CVPR, EMNLP
- Follow relevant social media feeds: e.g., public AI leaders
- Follow leading organizations, such as:
  - Epoch AI (https://epoch.ai/)
  - AI Future Project (https://www.aifutures.org/)
  - Anthropic (https://www.anthropic.com/research)
- Follow AI news sources; e.g., There's an AI For That (https://theresanaiforthat.com/)

Get Involved at CU Boulder

- Apply to join research efforts… see Canvas announcements

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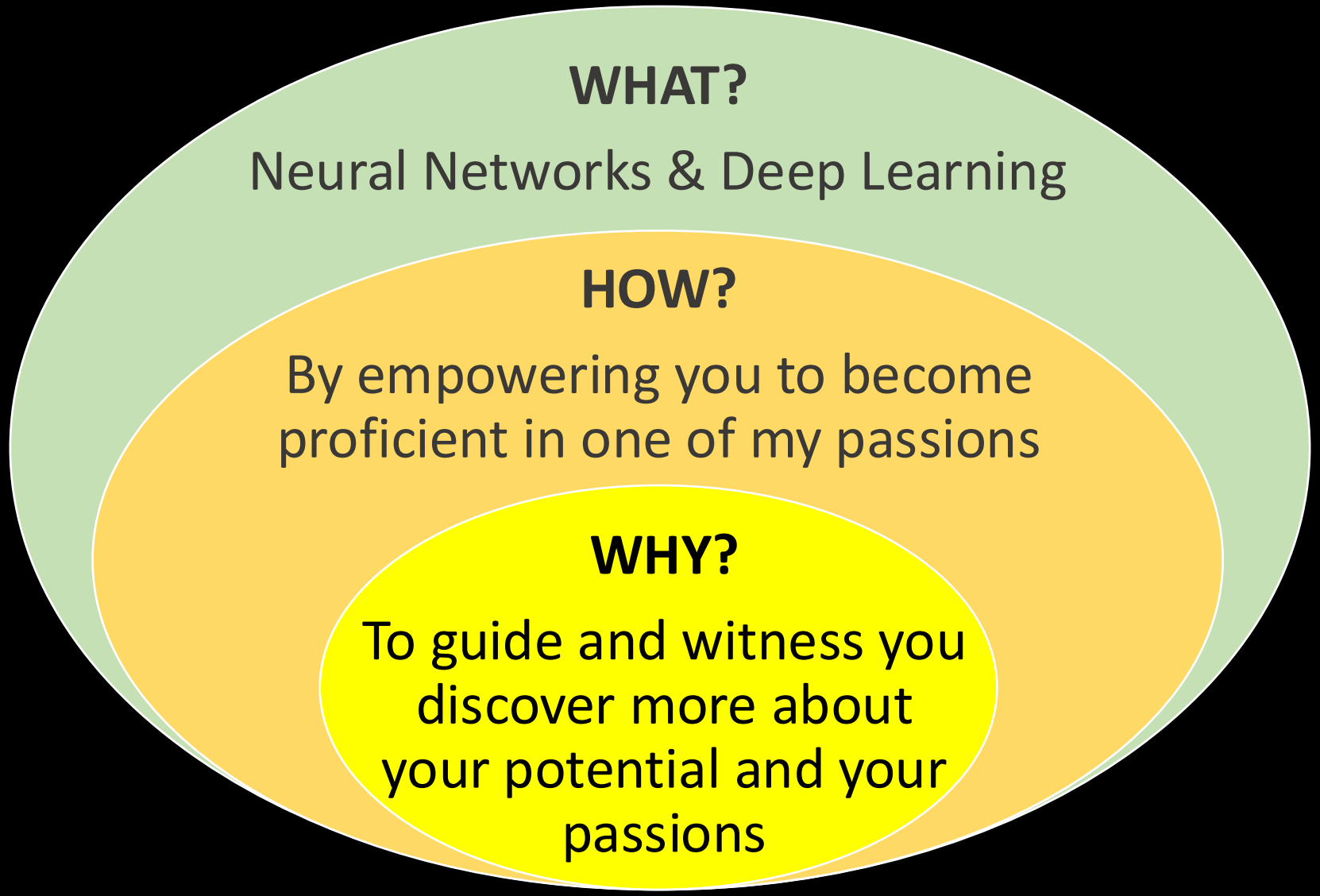
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Any questions for me?

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Recall My “Why” for Teaching You…



It’s been a pleasure teaching each of you.

Thank you for choosing to take this course.

Feel free to stay connected via LinkedIn ☺

The End